

lecture 3:-

$$\frac{B \times 12}{P11} :-$$

You can access the Internet only if
you are a CS major or
you are not a freshman.

Let p_2 you can access the Internet.
 q_2 you are a CS major.
 r_2 you are a freshman.

$$p \rightarrow (q \vee \neg x)$$

Ex 13 Hw.
p 11.

Propositional logic.

- System Consistency

- Puzzles

- Bodhan Seadhy.

$$\frac{12 \times 15}{12}$$

Bx15
P12. "The diagnostic msg is stored in buffer (00
p12. It is transmitted"

TP "the diagnostic msg is not stored in buffer"

$p \rightarrow a^0$ - then the diagnostic msg is stored in buffer
- then it is set transmitted".

Let $\overbrace{p^2}^{q^2}$ the diagnostic msg is stored in buffer.
" " " sets transmitted "

$$p \vee q \rightarrow T \quad \text{--- (1)}$$

$$TP = T \quad \text{--- (2)}$$

$n \rightarrow a.$ τ . T $\underbrace{\quad}_2$

$\downarrow$
 $P \quad \neg P$
 ~~$P \quad \neg P$~~
 $(P) \quad T$

$\downarrow$
 $P \quad q \quad P \vee q$
 ~~$P \quad q \quad P \vee q$~~
 ~~$P \quad q \quad P \vee q$~~
 $\rightarrow P \quad (T) \quad T$

$P \rightarrow q \quad z \quad T \quad \text{--- (2)}$
 $P \rightarrow q \quad z \quad T \quad \text{--- (3)}$

$(P) \quad T \quad \text{--- (2)}$
 $\rightarrow P \quad (T) \quad T$
 ~~$P \rightarrow T \quad z \quad T$~~

from (2) $P \rightarrow P \quad \text{--- (4)}$

from (1) put (4) $q \rightarrow T \quad \text{--- (5)}$

Constant.

from (3) put (4,5) $P \rightarrow T \quad z \quad T$
 $T \rightarrow T \quad \text{holds.}$

Ex 16
 P12. "The diagnostic msg is stored in buffer (or)
 $P \vee q$ it is retransmitted"

$\neg P$ "the diagnostic msg is (not) stored in buffer"

$P \rightarrow q$ "if - then the diagnostic msg is stored in buffer
 it is retransmitted".

$\neg q$ "the diagnostic msg is not retransmitted".

Let $P \rightarrow$ the diagnostic msg is stored in buffer.
 $q \rightarrow$ " " " " " retransmitted".

$P \vee q \quad z \quad T \quad \text{--- (1)}$

$\neg P \quad z \quad T \quad \text{--- (2)}$

$P \rightarrow q \quad z \quad T \quad \text{--- (3)}$

$\neg q \quad z \quad T \quad \text{--- (4)}$

from (2) $P \rightarrow P \quad \text{--- (4)}$

from (4) $q \rightarrow P \quad \text{--- (5)}$

put (4,5) in (1)

$P \vee P \quad z \quad T$

$P \neq T$

Inconsistent -

System Consistency. $\rightarrow$ Consistent -
 $\rightarrow$ Inconsistent -
 $\rightarrow$ Inconclusive -

$$p \rightarrow q = T$$
$$p \vee q = T$$
$$q \rightarrow p = T$$

Assignment 1:-

Do Ex 47-52.
Page 19-20.

8th - Sep - 2025.

- Handwritten. - put in box. - don't know.

Ques #1.

4th - Sep - 2025.

Determine if the system given by the statements below is Consistent / Inconsistent / Inconclusive.

$$P \wedge Q$$

$$\neg Q \vee R$$

$$P \rightarrow R$$

$$\neg P$$

Solution:-

$$P \wedge Q = T \quad \text{--- (1)}$$

$$\neg Q \vee R = T \quad \text{--- (2)}$$

$$P \rightarrow R = T \quad \text{--- (3)}$$

$$\text{from (4) } P = R \quad \text{--- (5)}$$

put (5) in (1).

$$P \wedge Q = T$$

$$= T$$

$$P \rightarrow R \quad \text{zT} \quad \text{--- (3)}$$

$$P \wedge Q = T.$$

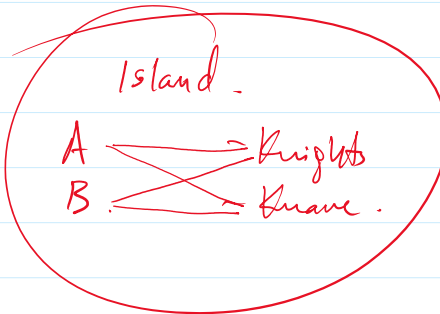
$$P \neq T.$$

$$\neg P \quad \text{zT} \quad \text{--- (4)}$$

Inconsistent.

- Puzzles.

الجزيرة



Knight always truth.
Knave " lies.

Ex 18 :-

P13

A Says " B is a Knight "

B Says " The two of us are of opposite type ".

Determine the type of A & B.

Let

$p \text{ z } A \text{ is a knight. } \neg p \text{ z } A \text{ is a knave}$

$q \text{ z } B \text{ is a knight } \neg q \text{ z } B \text{ is a knave}$

A	B.
Knight	Knight
Knight	knave
knave	knight
knave	knave.

A Says " B is a knight " = q .

B Says " A is a knight and B is a knave OR
A is a knave and B is knight ".

$$(p \wedge \neg q) \vee (\neg p \wedge q).$$

A	B.
Knight	Knight.

$$q = T$$

--- ①

$$p = T$$

$$\neg p = F$$

$$(p \wedge \neg q) \vee (\neg p \wedge q) = T \text{ --- ②.}$$

$$q = T$$

$$\neg q = F$$

from ① $T \neq T \checkmark$

from ② $(T \wedge P) \vee (P \wedge T) \neq T$

$$P \vee P \neq T$$

$$P \neq T$$

Case does not hold.

A

B.

Knight

Knave

$P = T$

$\neg P = F$

$q = T$

— ①

$q = F$

$\neg q = T$

$$(P \wedge \neg q) \vee (\neg P \wedge q) = F \text{ — ②}$$

from ① $P \neq T$

Does not hold.

Case 3

A

B.

Knave

Knight

$P = F$

$\neg P = T$

$q = T$

$\neg q = F$

$q = F$

— ①

$$(P \wedge \neg q) \vee (\neg P \wedge q) = T \text{ — ②}$$

from ① $T \neq F$

Does not hold.

Case 4

A

B.

Knave

Knave

$q = F$

— ①

$$(P \wedge \neg q) \vee (\neg P \wedge q) = F \text{ — ②}$$

$P = F$

$\neg P = T$

$q = F$

$\neg q = T$

from ① $F \neq F \checkmark$

from ② $(F \wedge T) \vee (T \wedge F) \neq F$

$$F \vee F \neq F$$

$$F \neq F \checkmark$$

1-2K ✓

Conclusion A & B are both knights.

A	B.
Knight	Knight
Knight.	Knight.
Knight	Knight.
Knight	Knight.